

DarkCORE: Efficient low-light object detection via collaborative reflectance denoising and object-oriented feature enhancement

Xin Feng ^{a,b,c}, Jie Wang ^a, Yunlong Wang ^a, Junxian Zeng ^a, Di Ming ^{a,b,c,*}

^a School of Computer Science and Engineering, Chongqing University of Technology, 400054, Chongqing, China

^b Chongqing Key Laboratory of Trusted Human-Machine Interaction and Security/Safety for Intelligent Connected Vehicles, China

^c Chongqing Intelligent Finance Research Collaborative Innovation Center, China

ARTICLE INFO

Keywords:

Low-light object detection
Self-supervised denoising
Collaborative reflectance denoising
Object-oriented feature enhancement

ABSTRACT

Object detectors trained on high-quality datasets often suffer performance degradation under low-light conditions, mainly due to amplified noise and distorted textures introduced by conventional enhancement methods. Most existing approaches focus primarily on brightness and contrast adjustment, which are designed for human visual perception and often fail to meet the requirements of downstream detection tasks. In addition, their denoising techniques rely on single-branch constraints and often struggle to balance noise suppression with detail preservation under complex lighting conditions. In this paper, we introduce DarkCORE, a detection-oriented and efficient low-light object detection framework that integrates Collaborative Reflectance Denoising and Object-Oriented Feature Enhancement. By analyzing the reflectance and illumination components of low-light images, DarkCORE applies a multi-branch self-supervised denoising strategy to the reflectance component, leveraging inter-branch structural differences to generate auxiliary learning signals that effectively reduce noise while preserving object-level detail. Furthermore, DarkCORE enhances object-oriented features in the latent feature space for both reflectance and illumination, enabling end-to-end joint optimization with the detector. Extensive experiments on the ExDark, DarkFace, and LOD datasets show that DarkCORE outperforms state-of-the-art methods across diverse low-light scenarios. In particular, DarkCORE achieves up to a +0.6 mAP improvement over strong baselines while introducing only 20K additional parameters, underscoring its efficiency and adaptability.

1. Introduction

With rapid advances in deep learning and the availability of high-quality datasets such as MS COCO (Lin et al., 2014) and Pascal VOC (Everingham et al., 2010), object detection models have achieved remarkable progress. However, their performance in low-light environments remains suboptimal. Low-light images are often characterized by reduced visibility, color distortion, and severe noise caused by insufficient illumination, darkness, and sensor exposure limitations. These degradations obscure discriminative object cues and fine-grained textures, posing significant challenges for reliable object detection. In practical applications such as intelligent video surveillance, nighttime autonomous driving, and unmanned aerial vehicle (UAV) monitoring, low-light images are commonly captured by resource-constrained edge devices, where both detection accuracy and computational efficiency are critical requirements. Therefore, low-light object detection must simultaneously address robustness under complex illumination and efficiency under limited computational budgets, motivating the need for detection-oriented and lightweight enhancement frameworks.

Existing low-light object detection methods can be broadly categorized into two paradigms: two-stage frameworks and single-stage end-to-end frameworks. In two-stage approaches, an image enhancement network and an object detector are treated as independent modules. The enhancement network is typically pre-trained on paired low/normal-light datasets and aims to improve contrast, brightness, and color balance, producing visually pleasing images for human observation. The enhanced images are then fed into a detector that is trained separately, following the standard detection pipeline. However, such enhancement strategies are primarily designed for human visual perception rather than machine perception. Although the visual appearance of low-light images can be improved, these methods do not explicitly consider the preservation of discriminative object structures. As a result, brightness and contrast enhancement often amplifies noise and distorts fine-grained textures in object regions, which are critical for accurate detection. Several representative methods (Fu et al., 2023a; Jiang et al., 2021; Liang et al., 2023; Luo et al., 2025; Wei et al., 2018; Wu et al., 2022; Yang et al., 2023) follow this paradigm and exhibit degraded detection performance under complex low-light conditions, as illustrated

* Corresponding author.

E-mail address: diming@cqut.edu.cn (D. Ming).

<https://doi.org/10.1016/j.eswa.2026.132424>

Received 7 November 2025; Received in revised form 9 March 2026; Accepted 8 April 2026

Available online 14 April 2026

0957-4174/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

in Fig. 1 (left). Single-stage frameworks attempt to alleviate this issue by integrating the enhancement network and the detector into a unified end-to-end architecture. In this setting, the enhancement module is optimized to improve detection accuracy rather than visual quality. Despite this advantage, existing single-stage methods still tend to focus on brightness or contrast correction and largely overlook the inherent noise characteristics of real-world low-light images, as illustrated in Fig. 1 (left). Noise is a dominant degradation factor in low-light scenarios due to high sensor gain and limited exposure, yet it is rarely modeled explicitly in current low-light detection frameworks. Conventional denoising methods (Guo et al., 2019; Quan et al., 2020; Zhang et al., 2018; Zhao et al., 2018) typically rely on single-branch mappings and require paired clean reference images for supervision. Such assumptions are impractical in real-world applications, where paired clean ground-truth images required by enhancement pretraining are typically unavailable. Consequently, existing denoising strategies struggle to balance noise suppression and texture preservation in a detection-oriented manner, limiting their effectiveness for downstream object detection tasks.

Furthermore, beyond noise handling and learning paradigm limitations, the *computational efficiency* of low-light object detection remains a critical yet underexplored issue. While some enhancement approaches can improve image quality, they often do so at the cost of inference speed. Many methods (Wang et al., 2025; Yi et al., 2023; Zhou et al., 2023), such as those based on Laplacian pyramids that enhance low-frequency components while restoring high-frequency details, introduce substantial architectural complexity. Similar issues are observed in several end-to-end low-light detection frameworks (Cui et al., 2022; Qin et al., 2022; Yin et al., 2023), where heavily designed enhancement modules significantly increase computational overhead and slow down detection. Such complexity severely limits their practicality in real-world low-light scenarios, particularly for resource-constrained platforms and real-time applications, highlighting the necessity of *efficient and lightweight* low-light detection frameworks that balance enhancement effectiveness with operational efficiency.

To address these challenges, this paper proposes an efficient low-light object detection framework via Collaborative Reflectance Denoising with Object-oriented Feature Enhancement (DarkCORE), explicitly optimized for detection-oriented enhancement rather than human-centric image restoration, within an end-to-end learning paradigm. Based on Retinex decomposition (Land & McCann, 1971), the reflectance component preserves object-related high-frequency details but is highly susceptible to noise amplification during enhancement, which can severely degrade target texture clarity. To counteract this, we introduce two **multi-branch collaborative denoising strategies** for the reflectance component: (i) a multi-branch collaborative mechanism that leverages inter-branch symmetry consistency, enabling DarkCORE to model and suppress noise without relying on clean reference images; and (ii) a noise-aware selection constraint that mitigates texture blurring in noisy regions while maximizing enhancement potential. The synergistic integration of these constraints strikes an optimal balance between denoising and detail preservation, thereby significantly improving image quality and boosting downstream detection performance, as illustrated on the left in Fig. 1.

Inspired by the latent diffusion model (Rombach et al., 2022), DarkCORE jointly optimizes denoised reflectance and illumination within the latent feature space to enhance features critical for detection. We develop two lightweight networks: (i) the Reflectance Enhancement Network (RE-Net), which extracts and color-corrects reflectance features to preserve texture; and (ii) the Illumination Enhancement Network (IE-Net), which processes illumination features via adaptive contrast adjustment. This synergistic combination not only improves recognition accuracy but also enables real-time processing by avoiding upsampling operations, thereby maintaining detection quality.

We conducted extensive experiments on the ExDark (Loh & Chan, 2019), UG^2 + DarkFace (Yang et al., 2020), and LOD (Hong et al., 2021) datasets, achieving results that surpass current state-of-the-art methods.

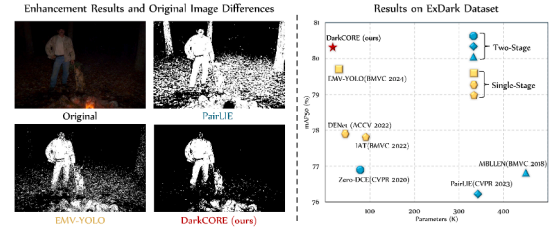


Fig. 1. Left: Enhancement differences between various methods and the original image, where white regions indicate areas of enhancement. Compared to PairLIE and EMV-YOLO, DarkCORE more effectively suppresses noise and enhances object-relevant features. Right: Comparison of parameter count and detection accuracy on the ExDark dataset.

Our approach also stands out for its parameter efficiency, with only 20K parameters, making it well suited for low-light object detection. As shown in Fig. 1 (right), the experimental results confirm the effectiveness of our method, delivering superior accuracy despite its minimal parameter count.

Our main contributions can be summarized as follows:

- We propose two multi-branch collaborative reflectance denoising strategies that enable efficient self-supervised denoising through multiple constraints, explicitly addressing the limitations of existing single-branch denoising methods in low-light detection.
- We design a target-feature enhancement network that integrates RE-Net and IE-Net to jointly optimize target features, effectively bridging the gap between low-light enhancement and downstream object detection.
- We design a lightweight enhancement network with only 20K parameters, striking an effective trade-off between detection accuracy and computational efficiency, while maintaining high detection accuracy and increasing feasibility for practical deployment.

2. Related work

Object Detection. Mainstream object detectors are generally categorized into two paradigms: single-stage and two-stage detectors. Single-stage detectors, such as SSD (Liu et al., 2016), YOLO (Redmon, 2018), and FCOS (Tian et al., 2019), directly predict object categories and bounding boxes in a unified forward pass, enabling fast inference. In contrast, two-stage detectors, including RCNN (Girshick et al., 2014), Faster R-CNN (Ren et al., 2015), and R-FCN (Dai et al., 2016), first generate region proposals and then perform classification and regression for refinement, typically achieving higher accuracy at the cost of efficiency. The rapid progress of object detection has been largely driven by large-scale annotated datasets such as COCO (Lin et al., 2014) and Open Images (Kuznetsova et al., 2020).

Low-Light Object Detection. Existing low-light object detection methods can be broadly divided into human-vision-oriented two-stage frameworks (Cai et al., 2023; Fu et al., 2023b; Guo et al., 2020; Yan et al., 2025; Zhang et al., 2019) and machine-vision-oriented single-stage frameworks (Cui et al., 2022, 2021; Du et al., 2024; Feng et al., 2024; Qin et al., 2022; Yin et al., 2023). The former typically relies on a pre-trained low-light image enhancement network to first restore images to a visually pleasing, well-lit appearance, followed by detector retraining. Representative approaches (Cai et al., 2023; Fu et al., 2023b; Wei et al., 2018; Yi et al., 2023; Yin et al., 2025) are often built upon Retinex theory (Land & McCann, 1971) and primarily aim to improve brightness, contrast, and color fidelity. However, these methods are mainly optimized for perceptual quality, and excessive enhancement often amplifies noise and distorts fine-grained textures, which are critical for object detection under low-light conditions.

In contrast, single-stage low-light detection methods (Cui et al., 2022; Du et al., 2024; Feng et al., 2024; Qin et al., 2022) jointly op-

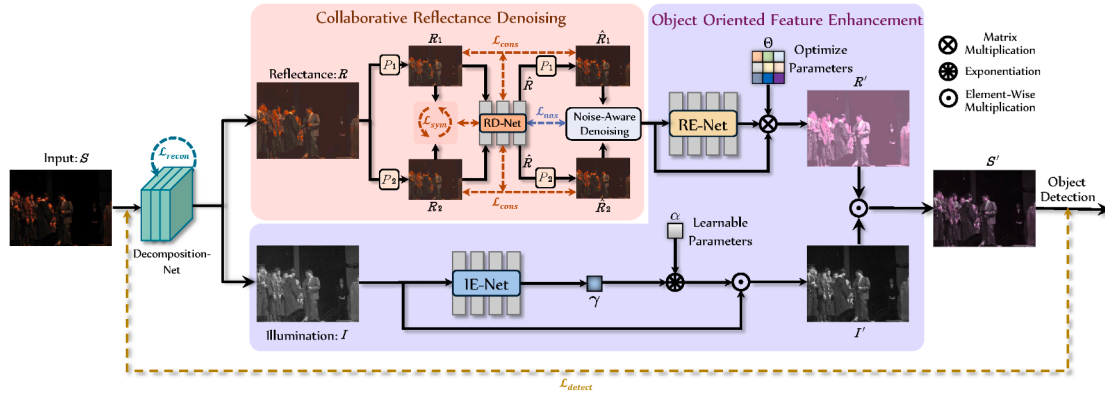


Fig. 2. Overview of the proposed DarkCORE framework. First, the input low-light image S is decomposed into reflectance R and illumination I by the Decomposition-Net. After that, the reflectance R is enhanced and denoised by RD-Net and further optimized by RE-Net, while the illumination I is enhanced by IE-Net to generate I' . Finally, the enhanced reflectance and illumination are fused to produce the final enhanced image S' , which is used for object detection. The entire framework is trained in an end-to-end manner with multi-level loss supervision.

timize the enhancement and detection modules with detection accuracy as the primary objective. For example, DAI-Net (Du et al., 2024) incorporates day–night domain adaptation into a Retinex-based framework, while EMV-YOLO (Feng et al., 2024) introduces semantic guidance to facilitate detection-oriented enhancement. Despite these advances, most existing single-stage approaches still lack explicit mechanisms to collaboratively model complex noise characteristics and preserve object-level textures during enhancement, leading to degraded detection performance in challenging low-light scenarios.

Self-Supervised Denoising. Noise2Noise (Lehtinen, 2018) enables self-supervised denoising by learning from pairs of independently corrupted observations, eliminating the need for clean reference images. ZS-N2N (Mansour & Heckel, 2023) further extends this paradigm to the single-image setting by constructing pseudo paired samples through downsampling, showing strong performance under unknown noise distributions. Our method is inspired by the noise-independence assumption adopted in ZS-N2N. However, existing self-supervised denoising approaches are primarily designed for human-oriented image restoration and typically optimize perceptual fidelity rather than task-level (e.g., object detection) performance. In contrast, we integrate self-supervised denoising into a Retinex-based enhancement framework and explicitly couple it with downstream object detection objectives, enabling effective noise suppression while preserving discriminative object-level structures that are critical for low-light detection.

3. Method

3.1. Preliminary

Inspired by the human visual system, the Retinex theory (Land & McCann, 1971) serves as an effective method for low-light image enhancement and has been widely applied in low-light object detection tasks. By decomposing an image into reflectance and illumination components, this theory enables separate processing of brightness and detail information, providing a solid theoretical foundation for low-light image enhancement.

Specifically, Retinex decomposition is mathematically expressed as $S = I \odot R$, where S represents the input low-light image, R denotes the three-channel reflectance component containing color information and high-frequency texture details, and I is the single-channel illumination component capturing brightness and low-frequency information. However, the decomposition process of I and R often lacks ground truth, posing challenges for training decomposition networks. To ensure that the original image can be accurately reconstructed from the decomposed I and R , it is necessary to impose constraints on the network. By incorporating the ℓ_2 -norm based reconstruction loss (Wei et al., 2018) to

optimize the decomposition process, the key visual characteristics of the original image can be preserved:

$$\mathcal{L}_{\text{recon}} = \|S - I \odot R\|^2, \quad (1)$$

where $\odot$ represents the Hadamard (element-wise) multiplication. As a result, I and R components can be better utilized for downstream tasks such as object detection.

3.2. Overview

To achieve efficient low-light object detection, we propose the DarkCORE model based on Collaborative Reflectance denoising and object-oriented feature Enhancement, as illustrated in Fig. 2. First, the low-light image S is decomposed into the reflectance component R and the illumination component I using the Retinex decomposition network. Second, in the collaborative reflectance denoising stage, the Reflectance Denoising Network (RD-Net) performs self-supervised denoising on R . By leveraging symmetric-consistency and noise-aware selection constraint, RD-Net achieves a precise balance between denoising effectiveness and detail preservation, producing a noise-free $\hat{R}$. Next, the object-oriented feature enhancement stage employs two lightweight networks, including the Reflectance Enhancement Network (RE-Net) and the Illumination Enhancement Network (IE-Net), to enhance the object semantic features in $\hat{R}$ and the I respectively. Then, the enhanced components R' and I' are combined to produce the final enhanced low-light image S' , which is subsequently fed into the object detector. By leveraging joint optimization in an end-to-end manner, DarkCORE achieves a dynamic balance among denoising, enhancement, and detection tasks, enabling efficient collaboration between components to enhance detection performance. We adopt a Retinex-based decomposition as it naturally decouples illumination and reflectance, allowing noise suppression and feature enhancement to be handled in a task-aware manner, which is more suitable for detection-oriented enhancement than single-branch alternatives. Unlike conventional Retinex-based pipelines where decomposition and denoising are often optimized primarily for visual restoration, our framework integrates reflectance denoising with downstream detection objectives through collaborative constraints and joint optimization. This task-aligned formulation ensures that enhancement operations are guided by detection performance, reducing the mismatch between perceptual restoration and recognition-oriented requirements.

3.3. Collaborative reflectance denoising

In low-light image processing, I is widely recognized for determining the image's dynamic range and is generally robust against noise interference. In contrast, R , which encapsulates the intrinsic properties of

the image such as high-frequency details, is more susceptible to noise during decomposition. This inherent issue in the Retinex decomposition process can be modeled as $S = I \odot R = I \odot (\hat{R} + N)$, where N represents the noise. Consequently, the noise contamination in R compromises the quality of the decomposition and negatively affects subsequent tasks.

Existing denoising methods, such as DnCNN (Zhao et al., 2018) and Noise2Noise (Lehtinen, 2018), are generally based on either a single network structure (e.g. CNN) or a single mapping paradigm (e.g. residual mapping). Due to the absence of multi-branch collaborative mechanism, these approaches fail to effectively characterizing the complex noise distribution. Inspired by the ZS-N2N method (Mansour & Heckel, 2023), we propose to construct two noisy observations from the reflectance component R , as shown in the top-left of Fig. 2. To be specific, a set of downsampling operations $P = \{P_1, P_2\}$ are utilized to transform $R \in \mathbb{R}^{H \times W \times C}$ into $R_i = P_i(R) \in \mathbb{R}^{H/2 \times W/2 \times C}$, for $i \in \{1, 2\}$. Regarding the downsampling operation, R is first divided into non-overlapping 2×2 pixel blocks. Subsequently, R_1 and R_2 are obtained by averaging the pixels along the diagonal and the anti-diagonal, respectively. This strategy follows common self-supervised denoising practices to construct complementary correlated observations, mainly serving a structural role.

Building upon this, the systematic exploitation of the mapping correlation between R_1 and R_2 enables a collaborative training mechanism for RD-Net without dependence on the clean ground truth image. Although originating from identical reflectance component R , R_1 and R_2 maintain partially independent noise characteristics. Thus, in the following, we propose two collaborative constraints to effectively harness the auxiliary supervised learning signals arising from this structural dissimilarity between R_1 and R_2 , thereby enhancing RD-Net's capability to learn a more robust denoising mapping relationship.

Symmetric Consistency Constraint. To overcome the limitations of existing single-branch approaches, we first introduce a multi-branch collaboration mechanism leveraging inter-branch symmetric consistency. By treating the denoised output of one branch as a supervised signal for the other, our method can enhance the RD-Net's ability to model complex noise distributions more effectively.

On the one hand, in the context of symmetry, we utilize the denoised reflectance information from one downsampling branch as the pseudo ground truth to train the noisy reflectance information from the other branch. Specifically, we design $f_N(\cdot)$ as a transformation network to extract noises from the noisy input. As a result, $R_i - f_N(R_i)$ can be utilized as the supervised signal to train R_j ($i \neq j$). Based on the symmetry of downsampling operation pair P_i and P_j in the set P , we propose the symmetry loss function to capture the noise distribution characteristics in the reflectance component as follows:

$$\mathcal{L}_{\text{sym}} = \|R_1 - (R_2 - f_N(R_2))\|^2 + \|R_2 - (R_1 - f_N(R_1))\|^2. \quad (2)$$

On the other hand, from the perspective of consistency, the denoised reflectance information before one downsampling branch is used as the pseudo ground truth to train the denoised reflectance information after the same downsampling branch. Thus, $P_i(R - f_N(R))$ can be utilized as the supervised signal to train $R_i - f_N(R_i)$. Considering the consistency between before and after downsampling operations P_i in the set P , the consistency loss function is introduced to constrain the error between the denoised result and the original input, as follows:

$$\begin{aligned} \mathcal{L}_{\text{cons}} = & \|R_1 - f_N(R_1) - P_1(R - f_N(R))\|^2 \\ & + \|R_2 - f_N(R_2) - P_2(R - f_N(R))\|^2. \end{aligned} \quad (3)$$

By integrating symmetry and consistency constraints, we construct a symmetric-consistency loss framework, where the total loss is defined as

$$\mathcal{L}_{\text{sycon}} = \mathcal{L}_{\text{sym}} + \mathcal{L}_{\text{cons}}. \quad (4)$$

This approach allows RD-Net to enhance denoising performance and improve the reconstruction of the reflectance component without the need

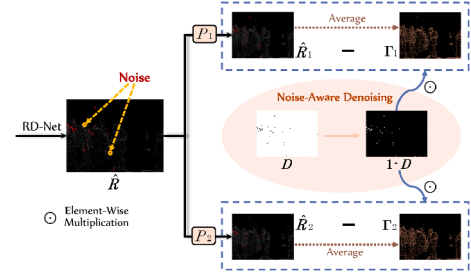


Fig. 3. Collaborative reflectance denoising strategy. Given a noisy reflectance image, multiple branches learn denoising features independently. A noise-aware selection module constrains the output based on spatial differences.

for any ground truth. From a theoretical perspective, the proposed multi-branch formulation introduces a symmetric-consistency mechanism to reduce the ambiguity inherent in self-supervised denoising, where multiple feasible solutions may satisfy a reconstruction objective. By introducing bidirectional inter-branch supervision, the proposed design constrains the feasible solution space of reflectance restoration and stabilizes optimization under complex low-light noise. This collaborative constraint further promotes the preservation of object-relevant structural information critical for downstream detection.

Noise-Aware Selection Constraint. Although the symmetric consistency loss improves RD-Net's denoising performance to some extent, relying solely on inter-branch supervision remains insufficient for capturing complex noise patterns. As shown on the left side of Fig. 3, the denoised reflectance component $\hat{R}$ (i.e., the output of RD-Net) still retains some noises, particularly in the regions with intricate textures or high-frequency details. Thus, in the following, we further introduce a noise-aware multi-branch collaboration mechanism to enable RD-Net to distinguish between “clean” and “noisy” regions, thereby precisely suppressing noises while preserving image details.

Firstly, we reapply the set P of downsampling operations to $\hat{R}$. Thus, we have complementary representations $\hat{R}_1 = P_1(\hat{R})$ and $\hat{R}_2 = P_2(\hat{R})$. Inspired by Zero-IG (Shi et al., 2024), we calculate the noise-aware difference D between $\hat{R}_1$ and $\hat{R}_2$ as follows:

$$D_{ij} = 2\sigma_1(i, j) \cdot \sigma_2(i, j) \cdot (\sigma_1(i, j)^2 + \sigma_2(i, j)^2 + K)^{-1}, \quad (5)$$

where $\sigma_1(i, j)$ and $\sigma_2(i, j)$ represent the standard deviation of the neighboring pixel's values for the (i, j) th pixel in $\hat{R}_1$ and $\hat{R}_2$ respectively, and K is a constant value added to stabilize the division for small denominators. In practice, K is set to a small positive value (e.g., $1e-5$) that mainly ensures numerical stability and has limited impact on the final detection performance. By definition, a higher D value indicates a smaller difference between $\hat{R}_1$ and $\hat{R}_2$. Based on this, we further refine D as a noise-aware binarized mask using a predefined threshold, i.e., $D = \Pi_\tau(D)$, where $\Pi_\tau(\cdot)$ denotes a thresholding operator defined as $\Pi_\tau(x) = \mathbb{I}(x \geq \tau)$ and $\mathbb{I}(\cdot)$ represents the indicator function. Consequently, $D = 1$ indicates clean regions, while $D = 0$ indicates noisy regions. The binarized mask D is then used in the subsequent loss formulation (Eq. 6), where the term $(1 - D)$ selects pixels belonging to noisy regions for denoising supervision. Details on the optimal value selection and a quantitative evaluation of the robustness of the proposed method with respect to the binary threshold are provided in Section 4.6. Moreover, due to the inherent correlation between adjacent pixels in clean images, the pixel values in $\hat{R}_i$ ($i \in \{1, 2\}$) are further refined by aligning them with the surrounding average. Specifically, we define $\Gamma_i[j, k]$ as the local average pixel value of $\hat{R}_i$ at the (j, k) th position, which enhances local features and guides each pixel to adjust toward the mean value of its surrounding 5×5 neighborhood window.

Similar to the symmetric consistency constraint, we employ the processed information Γ_i as the supervised signal to guide the preceding information $P_i(\hat{R})$. Consequently, minimizing the discrepancies of two branches facilitates collaborative training, refining the RD-Net's output

$\hat{R}$. Specifically, as illustrated in Fig. 3, leveraging the positional information of the noise-aware binarized mask D , we identify the nonzero elements in $1 - D$, effectively pinpointing white regions that require denoising. To further enhance the denoising performance, we introduce a noise-aware selection loss that guides the pixel values of $P_i(\hat{R})$ toward the neighborhood mean value Γ_i as follows:

$$\mathcal{L}_{\text{nas}} = \sum_{i=1}^{|P|} \left\| (1 - D) \odot (P_i(\hat{R}) - \Gamma_i) \right\|^2. \quad (6)$$

Finally, we define the collaborative denoising loss $\mathcal{L}_{\text{denoise}}$ as a weighted sum of the symmetric-consistency and noise-aware selection losses to train RD-Net:

$$\mathcal{L}_{\text{denoise}} = \lambda_{\text{sycon}} \mathcal{L}_{\text{sycon}} + \lambda_{\text{nas}} \mathcal{L}_{\text{nas}}, \quad (7)$$

where λ_{sycon} and λ_{nas} are weighting coefficients controlling the contribution of each term. By leveraging multiple constraints, RD-Net effectively performs self-supervised denoising under complex noise distributions, significantly improving the denoising performance and the reconstruction ability of the reflectance component. Compared to conventional single-branch reconstruction objectives, the proposed multi-constraint formulation provides stronger regularization and reduces the solution ambiguity in self-supervised optimization. This leads to more stable reflectance restoration and better preservation of object-sensitive features. A detailed analysis of the impact of these loss terms on detection performance is provided in Section 4.6.

3.4. Object-oriented feature enhancement

To enhance the efficiency and effectiveness of downstream tasks, such as object detection, we further propose the object-oriented feature enhancement method, which comprises RE-Net and IE-Net. Through the design of lightweight networks, the reflectance component $\hat{R}$ and illumination component I are mapped to a latent feature space, enabling the efficient extraction of high-level features relevant to object detection. This approach not only reduces computational complexity but also emphasizes semantic features and critical details of the objects, ultimately enhancing detection performance.

Reflectance Enhancement Network. RE-Net extracts and refines the features of the reflectance component $\hat{R}$ through deep convolutions, point-wise convolutions, and a four-stage down-sampling process. In the first two stages, channel and spatial structures are reorganized to effectively avoid additional computational overhead. In the last two stages, a maximum adaptive pooling method is employed to retain key features and unify feature dimensions. Subsequently, a color correction process is applied to produce a detail-rich and color-corrected reflectance component, i.e., $R' = \Theta \otimes \hat{R}$, where $\Theta = f_{\text{MLP}}(f_{\varepsilon_R}(\hat{R})) + E$. Here, E represents a learnable 3×3 matrix, $f_{\varepsilon_R}(\cdot)$ represents the feature extraction network, and $f_{\text{MLP}}(\cdot)$ represents the MLP network that maps the feature map into a feature matrix.

Illumination Enhancement Network. As shown in Fig. 2, IE-Net implements the illumination enhancement module in the proposed framework and realizes the illumination modulation formulated. IE-Net begins by converting the illumination component I to grayscale, highlighting edges and foreground features. To address the loss of original color information, we design a dual-path neural network $f_{\varepsilon_I}(\cdot)$ to enhance critical features, which employs an attention mechanism to extract comprehensive features and generate a contrast factor as $\gamma = f_{\varepsilon_I}(I)$, where $\gamma \in \mathbb{R}^{H \times W}$ is a spatially adaptive illumination adjustment map, and each element reflects the enhancement strength at the corresponding spatial location. Then, a learnable coefficient α is introduced to fine-tune γ , where α is implemented as a learnable scalar parameter that controls the non-linear strength of illumination enhancement, resulting in the adjusted expression as $I' = \gamma^\alpha \odot I$. This operation is performed within IE-Net and specifically targets illumination enhancement under low-light conditions. Here, I' represents the enhanced

illumination component. Finally, I' and R' are fused to generate the enhanced low-light image as $S' = I' \odot R'$, which serves as the final output of the enhancement module and is directly fed into the downstream object detection network.

3.5. Network training

During training, our DarkCORE model adopts a shared loss function framework with the detector, enabling end-to-end joint optimization. The Retinex decomposition task is guided by a reconstruction loss, while the reflectance denoising task utilizes a denoising loss to effectively suppress noise and avoid over-denoising.

As a result, the overall loss for our low-light object detection model is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{denoise}} + \mathcal{L}_{\text{detect}}, \quad (8)$$

where $\mathcal{L}_{\text{recon}}$ represents the Retinex reconstruction loss, $\mathcal{L}_{\text{denoise}}$ represents the reflectance denoising loss, and $\mathcal{L}_{\text{detect}}$ represents the YOLOv3 detection loss. This composite loss formulation follows the common paradigm in end-to-end low-light detection frameworks, where enhancement-related objectives are jointly optimized with the detection loss. All three loss components are seamlessly integrated into the MMDetection (Chen et al., 2019) framework. The overall loss $\mathcal{L}$ is then minimized using stochastic gradient descent (SGD) (Bottou, 1998), enabling end-to-end training of the network weights.

4. Experiments

We conducted two main experiments: low-light object detection (Section 4.1) and face detection (Section 4.2). Additional evaluations include generalization tests on different datasets and detectors (Sections 4.3-4.4), mAP threshold analysis (Section 4.5), ablation and loss-weight studies (Section 4.6), and efficiency analysis (Section 4.7). All experiments were conducted using the MMDetection (Chen et al., 2019) framework and were run on a workstation equipped with two NVIDIA RTX 3090 GPUs. Since our method focuses on detection-oriented enhancement under real low-light conditions, paired clean ground-truth images are unavailable, making traditional image quality metrics such as PSNR and SSIM inapplicable. Therefore, we adopt detection accuracy as the primary and task-relevant evaluation criterion throughout our experiments.

4.1. Object detection in darkness

Settings. All experiments were conducted under a unified and strictly controlled training protocol to ensure fair comparison. We employed the YOLOv3 detector with a Darknet-53 backbone, implemented based on the official MMDetection framework. All input images were resized to a fixed resolution of 608×608 during both training and inference. The detector was initialized with weights pre-trained on the COCO dataset (Lin et al., 2014) and fine-tuned on low-light datasets. We used the SGD optimizer with a momentum of 0.9 and a weight decay of $4e-5$. The batch size was set to 12, and the model was trained for 25 epochs. A linear warm-up strategy was applied during the first 3,000 iterations, gradually increasing the learning rate to $1e-3$. The learning rate was then decayed by a factor of 0.1 at the 6th, 15th, and 21st epochs. Unless otherwise specified, all comparative methods were trained using exactly the same detector architecture, input resolution, optimizer settings, data augmentation strategies, and training schedule. To reduce the influence of randomness, all experiments were repeated three times with different random seeds, and the reported results are averaged over these runs.

Datasets. We evaluated the proposed framework on the ExDark dataset (Loh & Chan, 2019), which contains 7,363 low-light images annotated with 12 object categories. Following the commonly adopted class-balanced split protocol, 80% of images from each category were used for training, and the remaining 20% were used for validation. To

Table 1

Detection performance comparison on the ExDark dataset. The table presents the mAP50 (%) results of various methods across 12 object categories under low-light conditions. (“bold”: best; “underline”: second-best).

Type	Method	Bicycle	Boat	Bottle	Bus	Car	Cat	Chair	Cup	Dog	Motorbike	People	Table	mAP50 (%)
Baseline	YOLOv3 (Redmon, 2018)	79.8	75.3	78.1	92.3	83.0	68.0	69.0	79.0	78.0	77.3	81.5	55.5	76.4
Two-stage	MBLEN (Lv et al., 2018) + YOLOv3	82.2	76.7	76.5	92.5	83.1	72.4	71.5	77.3	78.5	74.5	80.8	55.6	76.8
	KinD (Zhang et al., 2019) + YOLOv3	80.9	75.0	75.8	93.3	82.4	69.4	69.2	79.0	76.9	76.3	79.6	55.4	76.1
	Zero-DCE (Guo et al., 2020) + YOLOv3	84.1	77.6	78.3	93.1	83.7	70.3	69.8	77.6	77.4	76.3	81.0	53.6	76.9
	Retinexformer (Cai et al., 2023) + YOLOv3	82.0	80.1	80.9	91.3	83.1	70.8	70.3	76.9	75.8	75.4	80.8	57.8	77.1
	PairLIE (Fu et al., 2023b) + YOLOv3	82.5	76.7	76.4	91.6	82.9	71.0	70.0	76.9	78.9	72.6	79.9	55.3	76.2
	HVI (Yan et al., 2025) + YOLOv3	79.7	77.5	76.3	91.8	82.3	69.8	72.0	74.8	77.4	77.2	79.5	54.3	76.1
Single-stage	MAET (Cui et al., 2021)	83.1	78.5	75.6	92.9	83.1	73.4	71.3	79.0	79.8	77.2	81.1	57.0	77.7
	IAT (Cui et al., 2022)	79.8	76.9	78.6	92.5	83.8	73.6	72.4	78.6	79.0	<u>79.0</u>	81.1	57.7	77.8
	DENet (Qin et al., 2022)	80.9	79.2	<u>80.1</u>	90.7	<u>84.5</u>	70.7	72.0	79.3	80.1	76.7	82.4	58.0	77.9
	PE-YOLO (Yin et al., 2023)	84.7	79.2	79.3	92.5	83.9	71.5	71.7	79.7	79.7	77.3	81.8	55.3	78.0
	DAI-Net (Du et al., 2024)	83.8	75.8	75.1	94.2	84.1	<u>74.9</u>	73.1	79.2	82.2	76.4	80.7	<u>59.8</u>	78.3
	EMV-YOLO (Feng et al., 2024)	82.8	<u>79.7</u>	79.8	<u>94.1</u>	84.7	<u>74.3</u>	<u>74.1</u>	83.1	<u>82.7</u>	78.1	83.6	59.3	<u>79.7</u>
	DarkCORE (ours)	<u>84.2</u>	80.1	77.4	93.1	84.4	75.2	76.5	<u>81.0</u>	84.4	81.8	84.0	61.1	80.3

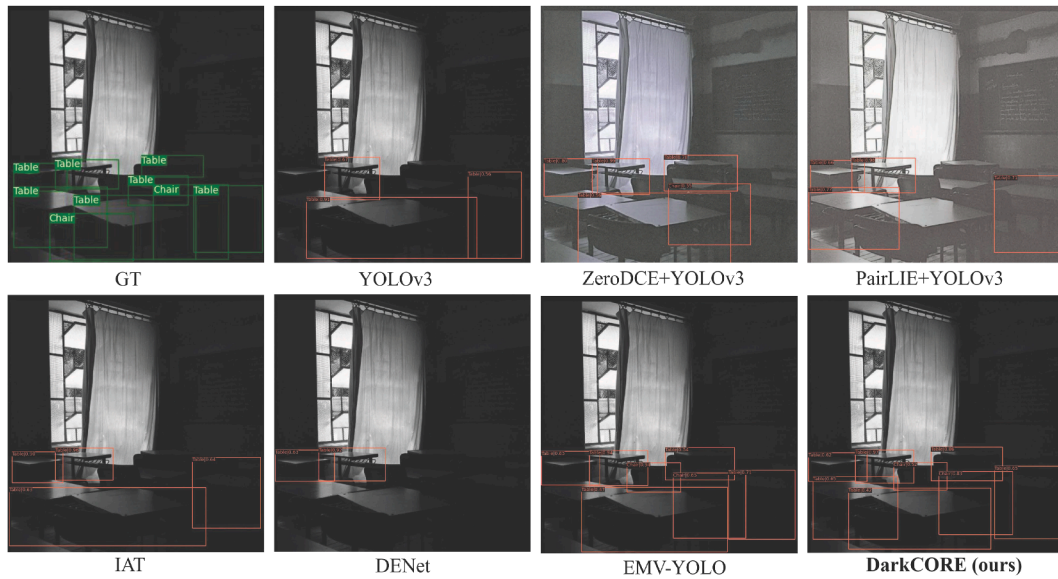


Fig. 4. Visual comparison of enhancement and detection results on the ExDark dataset. The first column shows the original low-light image, followed by results from baseline methods. The final column shows the output of our method.

ensure fair comparison, we strictly used the same data split for all compared methods. No images from the validation set were used during training, and no overlap exists between the training and validation sets.

The experimental results are summarized in Table 1, with corresponding visualizations shown in Fig. 4. Specifically: (i) For low-light image enhancement models based on human visual recovery, simply adding the YOLOv3 detector does not consistently improve detection performance in low-light scenes. In some cases, it may even lead to a slight performance degradation. For example, the detection accuracy of KinD (Zhang et al., 2019), PairLIE (Fu et al., 2023b), and HVI (Yan et al., 2025) shows marginal decreases compared to the YOLO baseline. This phenomenon highlights the mismatch between optimization objectives of image restoration and object detection tasks. (ii) For integrated enhancement methods and domain generalization approaches, the YOLO detector generally benefits from task-aware low-light processing. For instance, PE-YOLO (Yin et al., 2023) achieved a 1.6 percentage point improvement in mAP50 over the YOLO baseline, demonstrating competitive performance. In contrast, our proposed DarkCORE does not rely on additional dark-scene pretraining and achieves the highest mAP under the same training protocol, benefiting from joint end-to-end optimization of the denoising and enhancement processes. Across multiple independent runs, DarkCORE achieves an average mAP of 80.1 ± 0.18

on ExDark, with the best reported result (80.3) lying within one standard deviation. This indicates that the observed improvement over EMV-YOLO (Feng et al., 2024) is consistent and reproducible rather than attributable to random training fluctuations. Although the absolute gain is moderate, it is achieved without introducing additional computational overhead, highlighting the practical value of the proposed enhancement module as a lightweight and plug-and-play solution that can be readily integrated into existing detection pipelines. Overall, these results demonstrate that explicitly coupling denoising and enhancement with detection objectives is crucial for achieving stable performance gains under real-world low-light conditions.

4.2. Face detection in darkness

Settings. The training configuration for face detection followed the same protocol as described in Section 4.1, including the detector architecture, input resolution, optimizer, batch size, and data augmentation strategies. The only difference lies in the learning rate decay schedule, where the learning rate was reduced by a factor of 0.1 at the 14th and 20th epochs to better adapt to the face detection task.

Datasets. We evaluated our method on the UG² + DarkFace (Yang et al., 2020), which contains 6,000 low-light face images. Following the partition strategy in prior work (Cui et al., 2021), we used 5,400

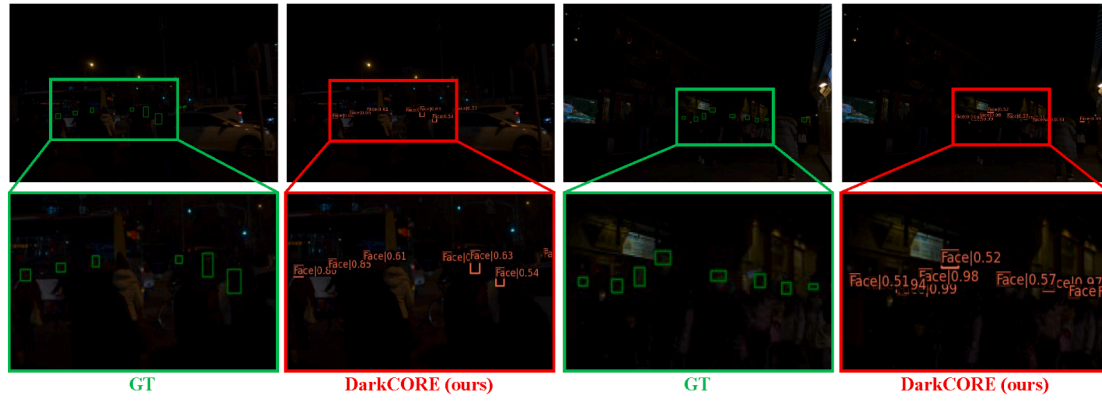


Fig. 5. Visualization results of DarkCORE on the UG² + DarkFace dataset, showing accurate face localization and effective noise suppression in challenging low-light scenes.

Table 2

Detection performance on the UG² + DarkFace dataset under low-light conditions. (“bold”: best; “underline”: second-best).

Type	Method	mAP50 (%)
Baseline	YOLOv3 (Redmon, 2018)	48.3
Two-stage	MBLEN (Lv et al., 2018) + YOLOv3	51.6
	KinD (Zhang et al., 2019) + YOLOv3	51.6
	Zero-DCE (Guo et al., 2020) + YOLOv3	54.2
	Retinexformer (Cai et al., 2023) + YOLOv3	57.4
	PairLIE (Fu et al., 2023b) + YOLOv3	55.4
	HVI (Yan et al., 2025) + YOLOv3	57.0
Single-stage	MAET (Cui et al., 2021)	55.8
	IAT (Cui et al., 2022)	53.1
	DENet (Qin et al., 2022)	51.2
	PE-YOLO (Yin et al., 2023)	51.1
	DAI-Net (Du et al., 2024)	57.0
	EMV-YOLO (Feng et al., 2024)	57.6
	DarkCORE (ours)	58.2

images for training and 600 images for testing. The same data split was consistently used for all compared methods.

Similar comparative experiments were conducted on the UG² + DarkFace (Yang et al., 2020) dataset to validate the generalization capability and detection accuracy of DarkCORE across different datasets. We compared DarkCORE with the same methods described in Section 4.1. As shown in Table 2, our method demonstrates superior performance, achieving a 0.6 percentage point improvement over the state-of-the-art EMV-YOLO (Feng et al., 2024). Across multiple independent training runs, DarkCORE achieves an average mAP of 57.8 ± 0.4 , indicating stable performance variations. Despite the relatively modest improvement, the proposed enhancement module preserves the original detector architecture and introduces negligible computational overhead, underscoring its modular lightweight design and compatibility with existing detection frameworks. Additional visualization results are presented in Fig. 5. These results indicate that the performance gain is consistently observed across runs and is not attributable to random effects during training.

4.3. Object detection on the LOD dataset

To comprehensively evaluate the effectiveness of the proposed method, we tested DarkCORE alongside several methods described in Section 4.1 on the LOD dataset (Hong et al., 2021). The LOD dataset is specifically designed for low-light object detection tasks, comprising 8 categories and 2,230 images that span various complex scenarios. For the experiments, the dataset was divided into a training set and a validation set using an 8:2 ratio by category. This setup followed the experi-

mental protocol outlined in Section 4.1 to ensure fairness and reliability in the results.

The experimental results (Table 3) show that DarkCORE improves detection accuracy by 1.0% compared to EMV-YOLO (Feng et al., 2024) on low-light object detection tasks. Given the limited size and challenging nature of the LOD dataset, even a 1.0% improvement indicates enhanced robustness under complex illumination conditions. This performance gain is attributed to the collaborative denoising and feature-oriented enhancement strategies, which effectively preserve details and enhance feature representations in low-light images—thereby improving the model’s adaptability to complex scenes. Additionally, the visual results in Fig. 6 demonstrate that DarkCORE achieves more precise target localization and boundary fitting, further validating its superior performance in low-light environments.

4.4. Low-light object detection using DETR

To examine the applicability of the proposed enhancement strategy under a different detection paradigm, we replace the YOLOv3 (Redmon, 2018) detector with DETR (Carion et al., 2020) and conduct experiments on the ExDark (Loh & Chan, 2019) dataset. This experiment is included as a supplementary evaluation and is not intended to serve as a direct performance comparison with YOLOv3-based results.

As shown in Table 4, DarkCORE achieves competitive detection accuracy when integrated with the DETR framework. Compared with EMV-YOLO (Feng et al., 2024), whose performance degrades noticeably when transferred from YOLO to DETR, our method exhibits a more stable performance trend under the DETR setting. Notably, the proposed enhancement module operates independently of the detector-specific prediction head or anchor design, as it primarily refines the reflectance representation before feature extraction. Therefore, its effectiveness does not rely on a particular detection architecture, but rather on improving input-level feature quality in a task-aware manner. These results suggest that the proposed framework maintains compatibility across different detection paradigms, including both anchor-based and transformer-based detectors.

4.5. mAP metrics and evaluation scope

To provide a more comprehensive evaluation of the model’s performance under different detection precision requirements, we additionally report both mAP@0.75 and mAP@[0.5:0.95], as shown in Table 5. The methods included for comparison are drawn from representative approaches discussed in Section 4.1. Our proposed DarkCORE consistently achieves the highest performance across two benchmark datasets: ExDark and UG² + DarkFace. Specifically, compared with the state-of-the-art EMV-YOLO (Feng et al., 2024), DarkCORE improves by 0.4 and 1.2

Table 3

Detection performance comparison of various methods on the LOD dataset. The best scores for each category and overall mAP50 (%) are highlighted in bold.

Method	Bicycle	Bottle	Bus	Car	Chair	Diningtable	Motorbike	Tvmonitor	mAP50 (%)
YOLOv3 (Redmon, 2018)	66.7	62.8	91.2	90.4	70.6	66.3	63.0	72.8	73.0
PairLIE (Fu et al., 2023b) + YOLOv3	66.6	66.1	89.3	90.5	70.4	65.1	68.9	64.3	72.6
HVI (Yan et al., 2025) + YOLOv3	66.7	63.2	92.3	89.8	71.4	69.4	67.1	72.7	74.1
IAT (Cui et al., 2022)	67.8	61.6	89.9	91.7	71.7	71.7	64.6	68.5	73.4
EMV-YOLO (Feng et al., 2024)	69.7	70.6	93.2	91.3	71.9	68.4	62.5	74.0	75.2
DarkCORE (ours)	70.0	70.9	89.8	91.4	72.0	70.8	67.7	77.2	76.2

Table 4

Detection performance comparison based on the DETR detector. Bold indicates the best result.

Method	mAP50 (%)
DETR (Carion et al., 2020)	74.7
PairLIE (Fu et al., 2023b) + DETR	73.3
IAT (Cui et al., 2022)	75.5
EMV-YOLO (Feng et al., 2024)	74.3
DarkCORE (ours)	76.1

Table 5

Detection mAP results at different IoU thresholds on ExDark and UG² + DarkFace datasets.

Method	ExDark		UG ² + DarkFace	
	mAP75	mAP[50:95]	mAP75	mAP[50:95]
YOLOv3 (Redmon, 2018)	43.2	42.4	7.7	17.9
PairLIE (Fu et al., 2023b)	43.8	42.6	12.1	21.1
HVI (Yan et al., 2025)	43.6	42.6	13.3	22.6
MAET (Cui et al., 2021)	46.6	44.4	11.8	21.6
EMV-YOLO (Feng et al., 2024)	47.8	44.8	11.9	21.9
DarkCORE (ours)	48.2	46.0	13.8	23.1

percentage points on ExDark, and by 1.9 and 1.2 points on DarkFace, in terms of mAP@0.75 and mAP@[0.5:0.95], respectively.

These results clearly demonstrate the effectiveness and robustness of our approach across diverse low-light scenarios. The consistent performance gains highlight the advantages of our design in bridging the gap between image enhancement and detection objectives, further verifying the superiority of our joint optimization strategy.

4.6. Ablation study

Ablation Study Setup. All ablation experiments in this section are conducted under a strict controlled-variable setting. Specifically, the detector architecture (YOLOv3), backbone network, input resolution, dataset partitioning, training schedule, optimizer configuration, and data augmentation strategies are kept completely identical to those used in the main experiments. The enhancement module is the only variable factor in all ablation studies unless otherwise specified. This design ensures that the observed performance differences can be directly attributed to the proposed enhancement and denoising mechanisms rather than variations in experimental configurations.

Reflectance Denoising Loss. We conducted a detailed evaluation of the loss function in RD-Net. The results indicate that combining the symmetric-consistency loss with the noise-aware selection loss significantly reduces noise and improves denoising performance. As shown in Table 6, omitting either loss term leads to a decline in performance. Notably, removing $\mathcal{L}_{\text{sycon}}$ results in the most substantial drop, demonstrating that $\mathcal{L}_{\text{sycon}}$ serves as a critical regularization constraint. It effectively preserves texture details during the denoising process and prevents over-smoothing. These findings validate the central role of the consistency loss in model optimization and underscore the importance of regularization in denoising tasks. Additionally, under the fixed detection pipeline described above, the degradation in denoising performance caused by

Table 6

Ablation study on the effects of different denoising loss components in RD-Net, evaluated on ExDark and UG² + DarkFace datasets.

Method	$\mathcal{L}_{\text{sycon}}$	$\mathcal{L}_{\text{nas}}$	ExDark	UG ² + DarkFace
YOLOv3	×	×	76.4	48.3
DarkCORE (ours)	✓	×	79.8	57.8
	×	✓	79.2	57.1
	✓	✓	80.3	58.2

Table 7

Ablation study evaluating the contribution of each module in our method. Performance is reported on ExDark and UG² + DarkFace datasets, along with the model parameter count.

Method	IE-Net	RD-Net	RE-Net	Parameters (K)	ExDark	UG ² + DarkFace
YOLOv3	×	×	×	×	76.4	48.3
DarkCORE (ours)	×	×	×	4.8	77.7	54.5
	✓	×	×	9.2	78.4	56.7
	×	×	✓	5.9	79.1	57.4
	×	✓	✓	16.1	79.8	57.9
	✓	✓	✓	20	80.3	58.2

removing either loss term also leads to a consistent drop in detection accuracy, indicating that effective reflectance denoising is crucial for downstream object detection.

Collaborative Denoising and Enhancement. All four configurations are evaluated using the same YOLOv3 detector under identical training settings, with the only variation being the enhancement strategy applied to the illumination and reflectance components. This experiment evaluates the effectiveness of the collaborative denoising and object-oriented enhancement modules. IE-Net is used to enhance the illumination component, while RD-Net and RE-Net are responsible for denoising and enhancing the reflectance component, respectively. We tested four configurations: (i) no enhancement applied to the original I and R components; (ii) enhancement applied to only the I or R component; (iii) denoising and enhancement applied only to the R component; and (iv) denoising and enhancement applied to both I and R .

The experimental results (Table 7) show that each module contributes to improved mAP50 (%) accuracy on both the ExDark and UG² + DarkFace datasets. As shown in Figs. 7 and 8, reflectance denoising and object-oriented feature enhancement effectively suppress noise, enhance target texture details, and maintain overall image quality—thereby providing high-quality input for downstream detection tasks. These results provide strong evidence that the proposed denoising and enhancement modules directly contribute to detection performance, as the detector architecture and training protocol remain fixed across all configurations.

Causal Ablation of Progressive Enhancement Stages. To analyze the causal contribution of each stage in our framework, we conduct a controlled ablation study under a fixed YOLOv3 detector. All experiments are performed using identical training settings, dataset splits, and optimization strategies, and only the enhancement pipeline is varied. We evaluate four progressive configurations following the actual processing

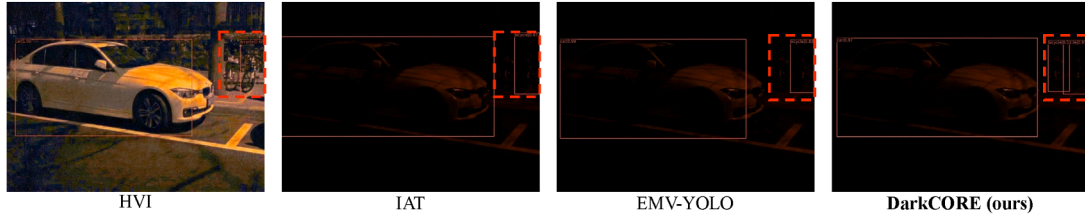


Fig. 6. Visualization of detection results from different methods on the LOD dataset. DarkCORE demonstrates more accurate localization and clearer object boundaries under complex low-light conditions.

Table 8

Ablation study of each enhancement stage under a fixed YOLOv3 detector. R, D, and E denote Retinex-based decomposition, reflectance denoising, and object-oriented feature enhancement, respectively.

Method	ExDark	UG ² + DarkFace
YOLOv3	76.4	48.3
YOLOv3 + R	78.2	53.3
YOLOv3 + R + D	79.8	57.8
YOLOv3 + R + D + E (Ours)	80.3	58.2

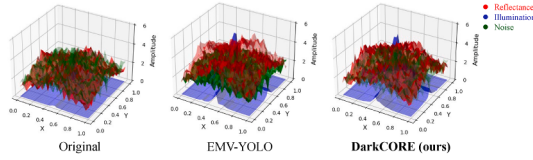


Fig. 7. Comparison of noise distributions in the reflectance, illumination, and noise components. The vertical axis (Amplitude) denotes the magnitude of pixel intensities in the decomposed reflectance, illumination, and noise maps, with values ranging from 0 to 6. Larger amplitude fluctuations indicate stronger noise variations in the image components. EMV-YOLO tends to amplify noise during enhancement, whereas DarkCORE suppresses noise while preserving image quality.



Fig. 8. Visualization of object-oriented feature enhancement. DarkCORE enhances object textures and boundaries while maintaining overall image clarity under low-light conditions.

flow of our method: (i) the baseline YOLOv3 detector; (ii) YOLOv3 with Retinex-based decomposition (R); (iii) YOLOv3 with Retinex decomposition and reflectance denoising (R + D); and (iv) the full model further incorporating object-oriented enhancement (R + D + E). As reported in Table 8, Retinex decomposition alone already improves detection accuracy by alleviating illumination interference. Adding reflectance denoising further boosts performance by suppressing noise while preserving object structures. The full model achieves the highest accuracy, demonstrating that object-oriented enhancement provides additional gains beyond decomposition and denoising. These results confirm a clear and consistent performance improvement across progressive enhancement stages, validating the independent and complementary contributions of each component under a unified detection setting.

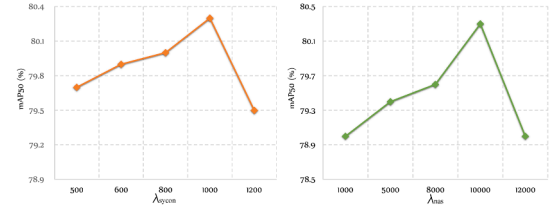


Fig. 9. Effect of adjusting the weighting coefficients for symmetry-consistency loss and noise-aware selection loss. Optimal detection accuracy is achieved when $\lambda_{\text{sycon}} = 10^3$ and $\lambda_{\text{nas}} = 10^4$.

Table 9

Sensitivity analysis of the binary noise-aware mask threshold in the proposed noise-aware selection constraint on the ExDark dataset.

Threshold	mAP50 (%)
0.95	79.6
0.965	79.9
0.97	80.1
0.975	80.3
0.98	79.8
0.99	79.7

Impact of $\mathcal{L}_{\text{sycon}}$ and $\mathcal{L}_{\text{nas}}$ on Detection. To more clearly examine the impact of different loss terms on detection performance and to avoid potential bias introduced by hyperparameter tuning across varying experimental settings, all experiments in this analysis are conducted using the same detector, training protocol, and dataset split. To assess the impact of the denoising loss weighting coefficients on detection performance, we conducted controlled experiments to determine the optimal configuration for the weighting parameters λ_{sycon} and λ_{nas} . Since the numerical magnitudes of the $\mathcal{L}_{\text{sycon}}$ and $\mathcal{L}_{\text{nas}}$ losses are considerably smaller than those of the primary detection task loss, their raw gradients may not effectively guide optimization in the denoising branch. Under consistent experimental conditions, we systematically varied the values of λ_{sycon} and λ_{nas} for comparative analysis. Specifically, when tuning one parameter, the other was fixed at its baseline value ($\lambda_{\text{sycon}} = 10^3$, $\lambda_{\text{nas}} = 10^4$). The experimental results, illustrated in Fig. 9, show that the model achieved its highest detection accuracy (mAP50 = 80.3) on the ExDark (Loh & Chan, 2019) dataset using $\lambda_{\text{sycon}} = 10^3$ and $\lambda_{\text{nas}} = 10^4$.

Effect of Binary Noise-Aware Mask Threshold. The binary noise-aware mask is introduced to selectively suppress noisy regions in the reflectance component, where excessive noise may lead to texture blurring and degraded object features. Specifically, the mask is generated by thresholding the estimated noise confidence map, where a higher threshold enforces stricter noise awareness and more aggressive noise suppression. Threshold values that are too low are excluded, as insufficient noise removal under such settings leads to degraded detection performance in low-light scenarios. Based on this consideration, we con-

Table 10

Efficiency comparison of various methods including parameters, runtime, and FLOPs. Highlighted values indicate the best (bold) and second-best (underline) performances.

Method	Parameters	Runtime (ms)	FLOPs (G)
MBLEN (Lv et al., 2018)	450K	38.1	81.98
KinD (Zhang et al., 2019)	8M	14.8	54.17
Zero-DCE (Guo et al., 2020)	79K	<u>4.7</u>	23.44
PairLIE (Fu et al., 2023b)	342K	13.6	100.91
IAT (Cui et al., 2022)	90K	13.8	6.48
DENet (Qin et al., 2022)	45K	3.3	1.42
PE-YOLO (Yin et al., 2023)	91K	32.2	8.11
EMV-YOLO (Feng et al., 2024)	27K	6.0	5.17
DarkCORE (ours)	20K	5.6	<u>4.34</u>

duct an ablation study by varying the threshold from 0.95 to 0.99 and report the corresponding detection performance on the ExDark dataset in terms of mAP@50. As shown in Table 9, the detection accuracy first increases as the threshold rises from 0.95 to 0.975, indicating that moderate noise-aware suppression helps prevent noise amplification while preserving discriminative object textures. However, further increasing the threshold beyond 0.975 leads to a performance drop, suggesting that overly strict masking may suppress useful fine-grained details together with noise, which is detrimental to detection. Based on this observation, we adopt a fixed threshold of 0.975 in all experiments, which provides the best trade-off between noise suppression and detail preservation under real-world low-light conditions.

4.7. Efficiency analysis

To evaluate parameter count, inference time (ms), and computational complexity (GFLOPs), we assessed the efficiency of each model on a single RTX 3090 GPU using three-channel images at a resolution of 544×544 pixels. Importantly, this resolution is used only for evaluating the computational efficiency of the enhancement modules, rather than for detection performance evaluation. All detection experiments in Sections 4.1–4.3 follow the standard protocol (Redmon, 2018) in low-light object detection and therefore use an input resolution of 608×608 to ensure fair comparison of detection accuracy. In contrast, the efficiency analysis aims to isolate the computational overhead introduced solely by the enhancement module, independent of the detector backbone and head. Following prior enhancement-based methods (Cui et al., 2022; Qin et al., 2022), we therefore evaluate the efficiency using a fixed input size of 544×544 . Since this evaluation focuses on the relative computational cost of the enhancement component, the exact spatial resolution does not affect the comparison but provides a consistent and controlled setting for measuring efficiency.

All efficiency metrics are reported under the same input resolution, detector configuration, and implementation environment to ensure fair comparison. Since all methods are based on the unmodified YOLOv3 (Redmon, 2018) architecture (except for MAET (Cui et al., 2021) and DAI-Net (Du et al., 2024)), we report parameter count and GFLOPs only for the enhancement modules while keeping the detector fixed. As shown in Table 10, DarkCORE achieves the lowest parameter count. Specifically, the additional parameters introduced by the proposed enhancement module account for only a small fraction of the overall detection framework, demonstrating its lightweight design. Moreover, it also exhibits competitive inference speed and low computational complexity, without modifying the detector backbone or prediction head, indicating that the proposed enhancement module introduces minimal overhead to the detection pipeline. This efficiency makes DarkCORE particularly suitable for real-time and resource-constrained low-light detection scenarios.

5. Conclusion

This paper presents DarkCORE, an efficient low-light object detection method that significantly enhances detection performance under low-light conditions. The method leverages Collaborative Reflectance denoising with object-oriented feature Enhancement within an end-to-end optimization framework. To address the challenges of noise suppression and feature preservation faced by traditional methods, we propose a multi-branch collaborative denoising strategy for the reflectance component, which effectively leverages inter-branch structural differences to generate auxiliary self-supervised learning signals. Additionally, our lightweight target-feature enhancement network-integrating RE-Net and IE-Net-optimizes object perception while generating high-quality inputs for detection tasks. With only 20K parameters, DarkCORE achieves fast inference and high accuracy. Extensive experiments validate its effectiveness and demonstrate its suitability for practical deployment. Overall, DarkCORE provides a robust framework for detection-oriented low-light enhancement, and future work will explore its adaptation to other detection architectures and video-based scenarios.

Funding

This work was supported by the Chongqing Municipal Education Commission Humanities and Social Sciences Research Base Project [grant number 24SKJD134]; the Natural Science Foundation of Chongqing, China [grant number CSTB2024NSCQ-MSX0341]; the National Natural Science Foundation of China [grant numbers 62441607, 62472059]; the Science and Technology Research Program of Chongqing Municipal Education Commission [grant number KJQN202301142]; the Open Research Fund of Key Laboratory of Cyberspace Big Data Intelligent Security, Ministry of Education [grant number CBDIS202403]; and the Chongqing University of Technology Postgraduate Innovation Fund Project [grant number gzlxc20243193].

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xin Feng reports financial support was provided by Chongqing Municipal Education Commission Humanities and Social Sciences Research Base Project. Di Ming reports financial support was provided by Natural Science Foundation of Chongqing, China. Di Ming reports financial support was provided by National Natural Science Foundation of China. Di Ming reports financial support was provided by Science and Technology Research Program of Chongqing Municipal Education Commission. Di Ming reports financial support was provided by Open Research Fund of Key Laboratory of Cyberspace Big Data Intelligent Security, Ministry of Education. Jie Wang reports financial support was provided by Chongqing University of Technology Postgraduate Innovation Fund Project. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Bottou, L. (1998). Online learning and stochastic approximations. *Online Learning in Neural Networks*, 17(9), 142.
- Cai, Y., Bian, H., Lin, J., Wang, H., Timofte, R., & Zhang, Y. (2023). Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 12504–12513).
- Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-end object detection with transformers. In *European conference on computer vision (ECCV)* (pp. 213–229).

- Chen, K., Wang, J., Pang, J., Cao, Y., Xiong, Y., Li, X., Sun, S., Feng, W., Liu, Z., Xu, J. et al. (2019). Mmdetection: Open mmlab detection toolbox and benchmark. *arXiv preprint arXiv:1906.07155*.
- Cui, Z., Li, K., Gu, L., Su, S., Gao, P., Jiang, Z., Qiao, Y., & Harada, T. (2022). You only need 90k parameters to adapt light: A lightweight transformer for image enhancement and exposure correction. *arXiv preprint arXiv:2205.14871*.
- Cui, Z., Qi, G.-J., Gu, L., You, S., Zhang, Z., & Harada, T. (2021). Multitask AET with orthogonal tangent regularity for dark object detection. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 2553–2562).
- Dai, J., Li, Y., He, K., & Sun, J. (2016). R-FCN: Object detection via region-based fully convolutional networks. *Advances in Neural Information Processing Systems*, 29, 379–387.
- Du, Z., Shi, M., & Deng, J. (2024). Boosting object detection with zero-shot day-night domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 12666–12676).
- Everingham, M., Gool, L. V., Williams, C. K. I., Winn, J., & Zisserman, A. (2010). The PASCAL visual object classes (VOC) challenge. *International Journal of Computer Vision*, 88, 303–338.
- Feng, X., Zeng, J., Wang, S., & He, Z. (2024). Toward highly efficient semantic-guided machine vision for low-light object detection. In *35th british machine vision conference* (pp. 25–28).
- Fu, H., Zheng, W., Meng, X., Wang, X., Wang, C., & Ma, H. (2023a). You do not need additional priors or regularizers in retinex-based low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 18125–18134).
- Fu, Z., Yang, Y., Tu, X., Huang, Y., Ding, X., & Ma, K.-K. (2023b). Learning a simple low-light image enhancer from paired low-light instances. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 22252–22261).
- Girshick, R., Donahue, J., Darrell, T., & Malik, J. (2014). Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 580–587).
- Guo, C., Li, C., Guo, J., Loy, C. C., Hou, J., Kwong, S., & Cong, R. (2020). Zero-reference deep curve estimation for low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 1780–1789).
- Guo, S., Yan, Z., Zhang, K., Zuo, W., & Zhang, L. (2019). Toward convolutional blind denoising of real photographs. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 1712–1722).
- Hong, Y., Wei, K., Chen, L., & Fu, Y. (2021). Crafting object detection in very low light. In *British machine vision conference (BMVC)* (p. 3). (vol. 1).
- Jiang, Y., Gong, X., Liu, D., Cheng, Y., Fang, C., Shen, X., Yang, J., Zhou, P., & Wang, Z. (2021). EnlightenGAN: Deep light enhancement without paired supervision. *IEEE Transactions on Image Processing*, 30, 2340–2349.
- Kuznetsova, A., Rom, H., Alldrin, N., Uijlings, J., Krasin, I., Pont-Tuset, J., Kamali, S., Popov, S., Mallocci, M., Kolesnikov, A. et al. (2020). The open images dataset v4: Unified image classification, object detection, and visual relationship detection at scale. *International Journal of Computer Vision*, 128(7), 1956–1981.
- Land, E. H., & McCann, J. J. (1971). Lightness and retinex theory. *JOSA*, 61(1), 1–11.
- Lehtinen, J. (2018). Noise2noise: Learning image restoration without clean data. *arXiv preprint arXiv:1803.04189*.
- Liang, X., Chen, X., Ren, K., Miao, X., Chen, Z., & Jin, Y. (2023). Low-light image enhancement via adaptive frequency decomposition network. *Scientific Reports*, 13(1), 14107.
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO: Common objects in context. In *European conference on computer vision (ECCV)* (pp. 740–755).
- Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.-Y., & Berg, A. C. (2016). Ssd: Single shot multibox detector. In *European conference on computer vision (ECCV)* (pp. 21–37).
- Loh, Y. P., & Chan, C. S. (2019). Getting to know low-light images with the exclusively dark dataset. *Computer Vision and Image Understanding*, 178, 30–42.
- Luo, H., Chen, B., Zhu, L., Chen, P., & Wang, S. (2025). Rcnnet: Deep recurrent collaborative network for multi-view low-light image enhancement. *IEEE Transactions on Multimedia*, 27, 2001–2014.
- Lv, F., Lu, F., Wu, J., & Lim, C. (2018). Mblen: Low-light image/video enhancement using cnns. In *British machine vision conference (BMVC)* (p. 4). (vol. 220).
- Mansour, Y., & Heckel, R. (2023). Zero-shot noise2noise: Efficient image denoising without any data. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 14018–14027).
- Qin, Q., Chang, K., Huang, M., & Li, G. (2022). Denet: Detection-driven enhancement network for object detection under adverse weather conditions. In *Proceedings of the asian conference on computer vision (ACCV)* (pp. 2813–2829).
- Quan, Y., Chen, M., Pang, T., & Ji, H. (2020). Self2self with dropout: Learning self-supervised denoising from single image. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 1890–1898).
- Redmon, J. (2018). Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*.
- Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster r-CNN: Towards real-time object detection with region proposal networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(6), 1137–1149.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 10684–10695).
- Shi, Y., Liu, D., Zhang, L., Tian, Y., Xia, X., & Fu, X. (2024). Zero-ig: Zero-shot illumination-guided joint denoising and adaptive enhancement for low-light images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 3015–3024).
- Tian, Z., Shen, C., Chen, H., & He, T. (2019). Fcos: Fully convolutional one-stage object detection. *arXiv preprint arXiv:1904.01355*.
- Wang, X., Yang, R., Wu, Z., Sun, L., Liu, J., & Zou, L. (2025). Ilenet: Illumination-modulated laplacian-pyramid enhancement network for low-light object detection. *Expert Systems with Applications*, 271, 126504.
- Wei, C., Wang, W., Yang, W., & Liu, J. (2018). Deep retinex decomposition for low-light enhancement. *arXiv preprint arXiv:1808.04560*.
- Wu, W., Weng, J., Zhang, P., Wang, X., Yang, W., & Jiang, J. (2022). Uretinex-net: Retinex-based deep unfolding network for low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 5901–5910).
- Yan, Q., Feng, Y., Zhang, C., Pang, G., Shi, K., Wu, P., Dong, W., Sun, J., & Zhang, Y. (2025). Hvi: A new color space for low-light image enhancement. In *Proceedings of the computer vision and pattern recognition conference (CVPR)* (pp. 5678–5687).
- Yang, W., Yuan, Y., Ren, W., Liu, J., Scheirer, W. J., Wang, Z., Zhang, T., Zhong, Q., Xie, D., Pu, S. et al. (2020). Advancing image understanding in poor visibility environments: A collective benchmark study. *IEEE Transactions on Image Processing*, 29, 5737–5752.
- Yang, X., Gong, J., Wu, L., Yang, Z., Shi, Y., & Nie, F. (2023). Reference-free low-light image enhancement by associating hierarchical wavelet representations. *Expert Systems with Applications*, 213, 118920.
- Yi, X., Xu, H., Zhang, H., Tang, L., & Ma, J. (2023). Diff-retinex: Rethinking low-light image enhancement with a generative diffusion model. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 12302–12311).
- Yin, L., Liu, P., Zhang, T., Zhao, H., & Zhou, Q. (2025). Differentiable histogram-guided unsupervised retinex enhancement for paired low-light images. *Expert Systems with Applications*, 298, 129782.
- Yin, X., Yu, Z., Fei, Z., Lv, W., & Gao, X. (2023). Pe-yolo: Pyramid enhancement network for dark object detection. In *International conference on artificial neural networks (ICANN)* (pp. 163–174).
- Zhang, K., Zuo, W., & Zhang, L. (2018). Ffdnet: Toward a fast and flexible solution for cnn-based image denoising. *IEEE Transactions on Image Processing*, 27(9), 4608–4622.
- Zhang, Y., Zhang, J., & Guo, X. (2019). Kindling the darkness: A practical low-light image enhancer. In *Proceedings of the 27th ACM international conference on multimedia* (pp. 1632–1640).
- Zhao, Y., Li, Y., Dong, X., & Yang, B. (2018). Low-frequency noise suppression method based on improved dnCNN in desert seismic data. *IEEE Geoscience and Remote Sensing Letters*, 16(5), 811–815.
- Zhou, D., Yang, Z., & Yang, Y. (2023). Pyramid diffusion models for low-light image enhancement. *arXiv preprint arXiv:2305.10028*.